ZOMATO DELIVERY ANALYTICS

Delivery Time Prediction & Operational Insights

PYTHON • SQL • MACHINE LEARNING • POWER BI

PROJECT AT A GLANCE

45,584	3.62 min	4.56 min	0.7638
Delivery records	Test MAE	Test RMSE	Test R^2

PROJECT OVERVIEW

This project examines delivery-time patterns in a food-delivery dataset and builds a machine-learning model to estimate how long an order may take. It combines data cleaning, exploratory analysis, SQL aggregation, model evaluation and an interactive Power BI dashboard. The aim is to turn raw delivery records into clear findings that an operations team could investigate further.

The report also documents a multiple-delivery what-if analysis. That analysis is explicitly labelled hypothetical: it estimates the result of an assumption and does not claim that a real routing or assignment optimisation system was built.

DELIVERABLES

- Cleaned dataset containing 45,584 records and 25 columns.
- Exploratory findings on traffic, weather, batching, festival status, vehicle condition and city.
- XGBoost regression model evaluated on a held-out test set.
- Power BI dashboard with KPI cards, category comparisons and slicers.
- Scenario comparison with assumptions and limitations.

Project title: Zomato Delivery Analytics & Time Prediction

01 | Executive Summary

Delivery time affects customer experience, planning and delivery expectations. This project analyses 45,584 records to understand how delivery time varies across recorded conditions and to test whether available order attributes can support prediction.

Python was used for data preparation and feature engineering, PostgreSQL for structured queries, XGBoost for prediction, and Power BI to present the findings. The model achieved a test MAE of 3.62 minutes, RMSE of 4.56 minutes and R^2 of 0.7638.

KEY FINDINGS

Finding	Observed result	Why it matters
Traffic	Jam: 31.18 min; Low: 21.46 min	Traffic category is associated with differences in delivery time.
Multiple deliveries	Group 0: 22.88 min; group 3: 47.82 min	Batch count is worth investigating in delivery planning.
Festival status	Festival: 45.52 min; No festival: 25.91 min	Festival periods may need separate analysis.
Model performance	MAE 3.62 min; R^2 0.7638	The model captures useful signal but still makes errors.

OBJECTIVES

- Prepare and inspect the dataset, handling invalid and missing values where appropriate.
- Compare delivery times across operational categories.
- Train and evaluate a regression model on a separate test set.
- Present results in a filterable dashboard.
- Explore a hypothetical policy scenario while being transparent about assumptions.

TOOLS AND THEIR ROLE

Python / pandas	PostgreSQL / SQL	XGBoost	Power BI
Cleaning, features, EDA	Queries and aggregation	Regression prediction	KPI cards and charts

02 | Dataset & Data Preparation

The source dataset contains 45,584 rows and 20 original columns. It records delivery-person attributes, restaurant and delivery coordinates, order timing, weather, traffic density, vehicle condition, order type, multiple deliveries, festival status, city and delivery duration. Five derived fields were added, giving a working dataset of 25 columns.

CLEANING AND FEATURE ENGINEERING

Step	Work performed	Reason
Validation	Reviewed missing values and invalid ratings and vehicle-condition values; invalid entries were treated as missing.	Reduce the influence of invalid values.
Imputation	Filled selected numeric fields with medians and categorical fields with modes.	Keep the dataset usable for analysis.
Standardisation	Changed “Metropolitian” to “Metropolitan”.	Make category labels consistent.
Date features	Converted order dates and derived month, day and weekday.	Allow temporal comparisons.
Distance	Calculated straight-line distance with Haversine and flagged implausible values.	Identify potentially unreliable coordinates.

IMPORTANT DATA-QUALITY NOTES

The distance calculated from latitude and longitude is a straight-line estimate, not a road-route distance. Geographic screening left 4,071 records without a usable screened distance. Extreme distance values were also present, so distance should be validated carefully before use in modelling.

For the reported model, distance, identifiers, raw coordinates and raw order-time fields were excluded from the feature list. The target variable was “Time_taken (min)”. The cleaned dataset’s mean delivery time was 26.29 minutes and its median was 26 minutes.

LIMITATIONS OF THE SOURCE DATA

Some categories are imbalanced. The Semi-Urban city group contains only 164 records, and festival orders are much less common than non-festival orders. A high average in a small group should be interpreted cautiously. Imputation also simplifies the data and can hide some original missingness patterns.

The dataset is observational. It records what happened in the available sample, but it does not by itself prove that a particular feature caused a delivery to be slower or faster.

03 | Exploratory Analysis & Business Insights

Exploratory analysis compared delivery time across traffic density, weather, multiple-delivery count, festival status, vehicle condition and city. These comparisons help identify patterns worth checking; they should not be interpreted as causal effects.

Factor	Group averages	Business interpretation
Traffic density	Low 21.46 min Medium 26.70 High 27.24 Jam 31.18	Jam traffic has the highest average. ETA estimates and staffing plans could consider traffic conditions.
Multiple deliveries	0: 22.88 min 1: 26.72 2: 40.45 3: 47.82	Higher batch counts coincide with longer times. Batch size and order sequence merit further study.
Weather	Sunny 21.86 min Cloudy 28.92 Fog 28.74	Weather categories differ, though location and traffic may also differ.
Festival status	Festival 45.52 min No festival 25.91	Festival periods may coincide with unusual demand or operating conditions.
Vehicle condition	0: 30.07 min 1: 24.43 2: 24.45	Verify category coding before drawing maintenance conclusions.
City	Metropolitan 27.14 Urban 22.98 Semi-Urban 49.73	Semi-Urban has only 164 records, so its average is less stable.

BUSINESS INTERPRETATION

Traffic-aware ETA messaging could be explored because the Jam group has a longer average delivery time than the Low group. Multiple-delivery count is another candidate for operational review, but the analysis does not establish that reducing batch size will automatically reduce every order's delivery time.

Festival and city comparisons should be checked alongside sample sizes, traffic and route characteristics. The same caution applies to weather: group averages describe the available data, not the isolated effect of weather.

QUESTIONS FOR FURTHER ANALYSIS

- Do patterns remain when comparing orders with similar distance and traffic conditions?
- Are model errors larger for festivals, jams or specific city categories?
- Do results change when data is split by date rather than randomly?
- Can a real policy be tested against a baseline with feasible routes and delivery constraints?

04 | Machine-Learning Model

A supervised regression model was built to estimate “Time_taken (min)”. The data was split into training and test sets with an 80/20 split and random_state=42. The training set contained 36,467 records and the test set contained 9,117 records.

MODEL CONFIGURATION

Component	Implementation
Algorithm	XGBoost regressor.
Parameters	300 estimators, learning rate 0.05, maximum depth 6, squared-error objective.
Categorical encoding	One-hot encoding with drop_first=True.
Inputs	Age and rating; weather; traffic density; vehicle condition; order type; vehicle type; multiple deliveries; festival; city; month; day; weekday.
Excluded fields	Identifiers, raw coordinates, raw order-time fields and distance were excluded from the reported feature list.

TEST-SET EVALUATION

Metric	Value	Interpretation
MAE	3.62 minutes	Average absolute difference between predicted and observed delivery time.
RMSE	4.56 minutes	Penalises larger errors more heavily than MAE.
R^2	0.7638	About 76.4% of target variance is explained on this test split.

FEATURE IMPORTANCE

The largest reported importance scores were low traffic density (0.197), multiple deliveries (0.130), sunny weather (0.108), delivery-person ratings (0.100) and festival status (0.095). These values show how the model used features; they are not causal effects and do not indicate how many minutes a feature independently adds or removes.

MODEL LIMITATIONS AND NEXT CHECKS

The evaluation uses a random hold-out split, so similar conditions may appear in both training and test sets. A time-based split would better reflect prediction on future orders. Useful next steps include comparing against a simple baseline, reviewing residuals, measuring error by traffic and city, and checking performance on rare categories.

05 | Power BI Dashboard

The dashboard brings the project's main indicators and comparisons into one view. It uses a dark navy canvas with red bars and accents, white labels and a yellow subtitle. Dropdown slicers let viewers filter by order date, traffic density, weather conditions and vehicle type.

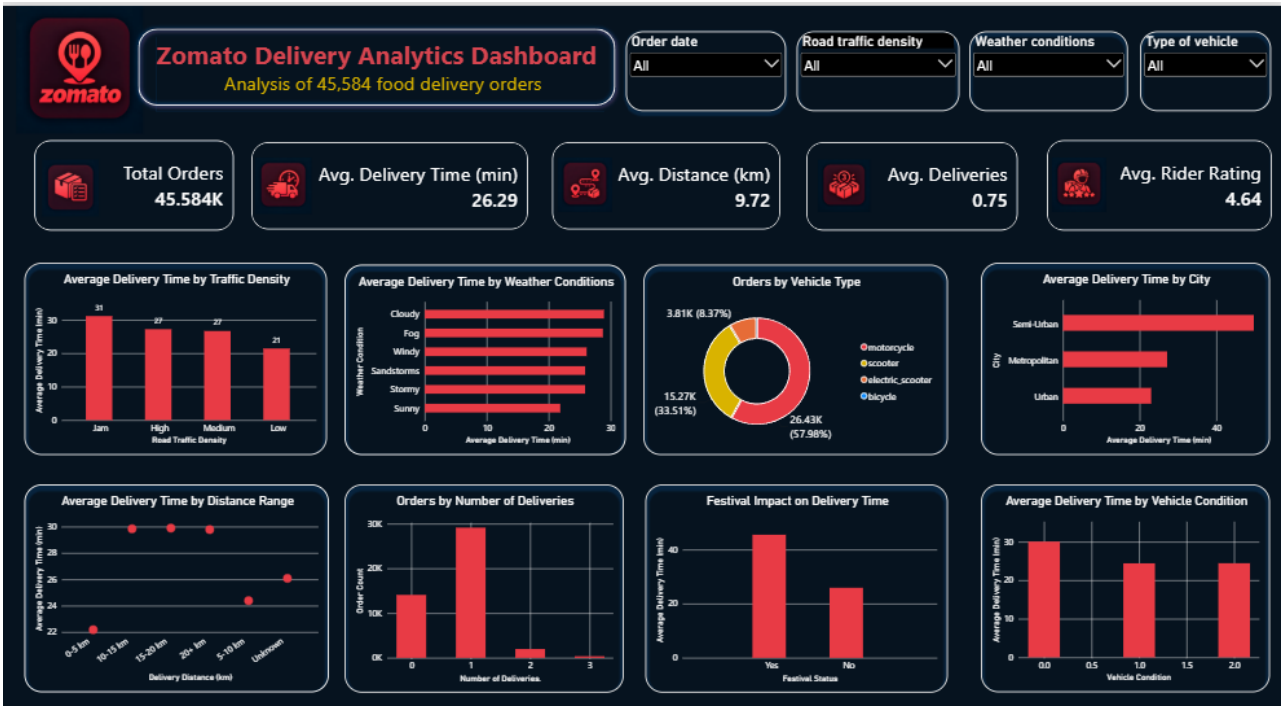


Figure 1. Power BI dashboard screenshot supplied for this report.

HEADLINE KPIs

Indicator	Value shown	Purpose
Total orders	45,584	Shows the volume of records represented.
Average delivery time	26.29 minutes	Provides a central service-time measure.
Average distance	9.72 km	Reproduced from the dashboard card.
Average deliveries	0.75	Summarises the multiple-delivery field as displayed.
Average rider rating	4.64	Shows the average rating measure as displayed.

The dashboard also compares average delivery time by traffic and weather; orders by vehicle type; delivery time by city and distance range; orders by multiple-delivery count; festival impact; and vehicle condition.

The KPI row provides a quick summary, while the charts explain how averages or counts differ by category. Slicers support focused exploration without requiring the viewer to inspect the raw table. Values are reproduced from the supplied screenshot; the report does not independently recalculate every card.

06 | Multiple-Delivery Scenario Analysis

Average delivery time was 40.45 minutes for orders with 2 multiple deliveries and 47.82 minutes for orders with 3. To explore a possible policy direction, a what-if scenario was created for selected orders in these two groups.

METHOD USED

The baseline average was 26.29 minutes. For each scenario, a reproducible random sample (random_state=42) was selected from records where multiple_deliveries was 2 or 3. A selected fraction of 10%, 20% or 30% was assigned the observed average time for the group with multiple_deliveries = 1, approximately 26.72 minutes. The overall mean was then recalculated.

Policy rate	Scenario average	Estimated saving / order	Estimated reduction
10%	26.22 min	0.07 min	0.28%
20%	26.14 min	0.15 min	0.58%
30%	26.07 min	0.23 min	0.87%

WHAT THE RESULTS MEAN

Under the stated assumption, the scenario mean falls as a larger fraction of selected orders is assigned the group-1 average. This is an arithmetic what-if calculation, not a validated prediction that delivery operations would achieve those savings. The output CSV, “Multiple_Delivery_Policy_Comparison.csv”, stores the scenario table so it can be inspected alongside the notebook.

CRITICAL LIMITATIONS

- No delivery routes or delivery-partner assignments were actually optimised.
- The group-1 mean is an observational benchmark and may not be attainable for selected orders.
- The scenario does not model route feasibility, order sequence, travel distance, capacity, customer time windows or rider workload.
- The small estimated reductions should not be presented as proven operational gains.

> **Correct wording: “The what-if analysis estimated a 0.28%–0.87% reduction under the stated assumption. Real-world savings were not measured.”**

07 | Conclusion & Future Scope

This project demonstrates an end-to-end analytics workflow on 45,584 food-delivery records. Python was used for data preparation, feature engineering and exploratory analysis; PostgreSQL supported structured aggregation; XGBoost was used for delivery-time prediction; and Power BI presented the main findings in an interactive dashboard.

The model achieved a test MAE of 3.62 minutes, RMSE of 4.56 minutes and R^2 of 0.7638. Descriptive analysis showed longer average times under jam traffic and for orders with higher multiple-delivery counts. These findings can guide further investigation into ETA communication, batch size and operating conditions, but they do not establish causality.

FUTURE SCOPE

Priority	Next improvement	Expected value
1	Use a time-based split and compare with a simple baseline model.	Stronger evidence of future prediction performance.
2	Review implausible coordinates and distance outliers before using distance as a feature.	More reliable geographic analysis.
3	Measure model error by traffic, city, vehicle type and multiple-delivery count.	Identify groups where predictions need improvement.
4	Build a constrained routing or assignment experiment using route distance, capacity and time windows.	Move toward testable operational optimisation.
5	Compare a proposed policy against a baseline on the same feasible test cases.	Quantify improvement more credibly.

FINAL TAKEAWAY

The strongest part of the current project is the combination of data analysis, a measurable prediction model and a clear dashboard. The multiple-delivery scenario adds evidence of analytical thinking, provided it remains labelled as hypothetical. The next meaningful upgrade is to test a feasible assignment or routing method rather than claim optimisation from a what-if substitution.

TOOLS USED | Python • pandas • XGBoost • PostgreSQL • SQL • Power BI

Prepared from the project metrics and dashboard image supplied by the project author. Figures and interpretations are limited to the documented workflow and supplied results.